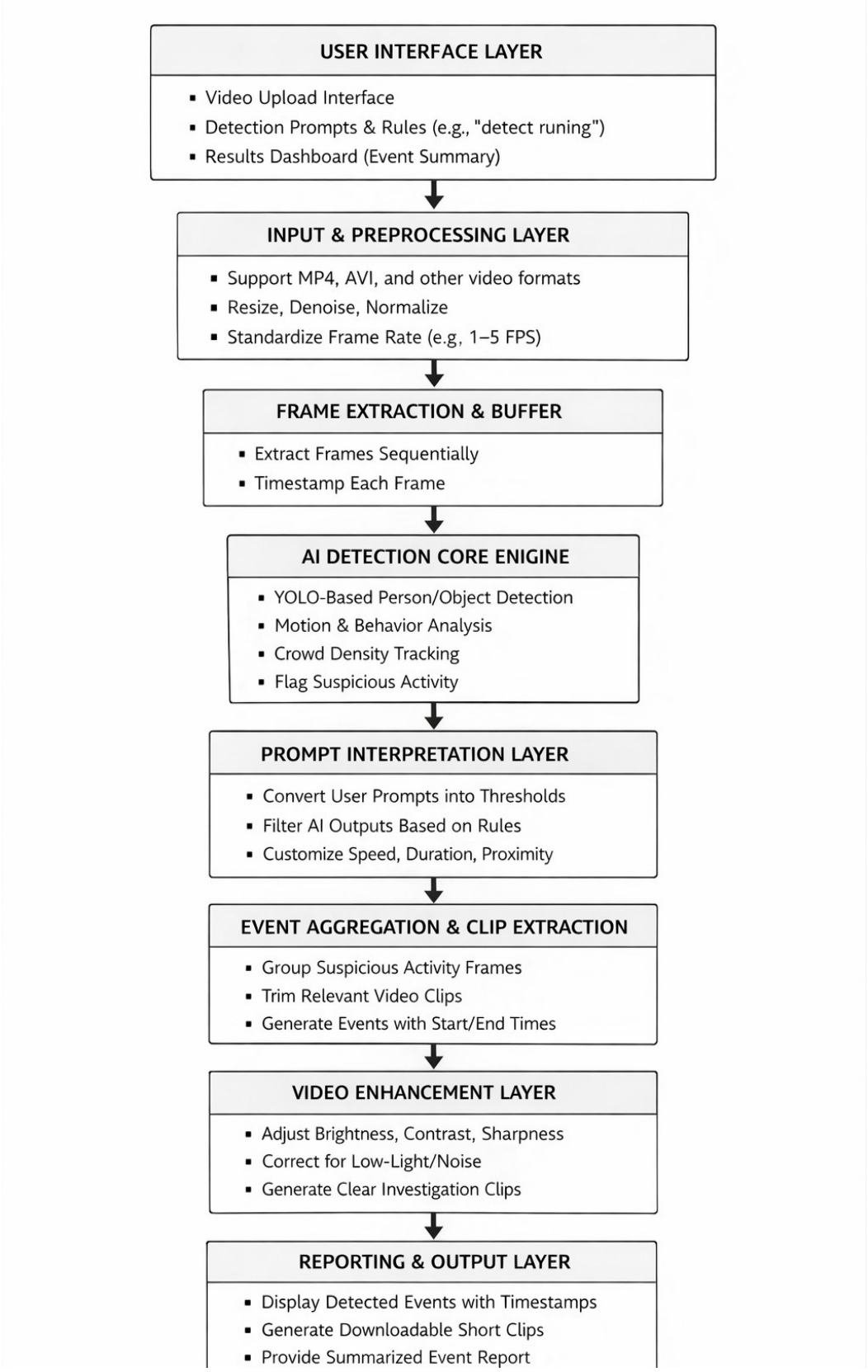

USER INTERFACE LAYER
Video Upload Interface
Detection Prompts & Rules (e.g., "detect runing")
Results Dashboard (Event Summary)
INPUT & PREPROCESSING LAYER
Support MP4, AVI, and other video formats
Resize, Denoise, Normalize
Standardize Frame Rate (e.g, 1–5 FPS)
FRAME EXTRACTION & BUFFER
Extract Frames Sequentially
Timestamp Each Frame
AI DETECTION CORE ENIGINE
YOLO-Based Person/Object Detection
Motion & Behavior Analysis
Crowd Density Tracking
Flag Suspicious Activity
PROMPT INTERPRETATION LAYER
Convert User Prompts into Thresholds
Filter AI Outputs Based on Rules
Customize Speed, Duration, Proximity
EVENT AGGREGATION & CLIP EXTRACTION
Group Suspicious Activity Frames
Trim Relevant Video Clips
Generate Events with Start/End Times
VIDEO ENHANCEMENT LAYER
Adjust Brightness, Contrast, Sharpness
Correct for Low-Light/Noise
Generate Clear Investigation Clips
REPORTING & OUTPUT LAYER
Display Detected Events with Timestamps
Generate Downloadable Short Clips
Provide Summarized Event Report